

Analyzing Ethereum-Based Electronic Health Records' Data Colonialism Ethical Contexts

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Abstract—This paper explores how Ethereum's unique features impact the ethical and landscape and data colonialist values surrounding Blockchains and electronic healthcare records (EHRs). Ethereum's unique features, like its energy-efficient proof-of-stake model, suggest potential advantages over a more general blockchain, of which many ethical analyses have been completed. These advantages save Ethereum from many ethical issues that blockchains such as Bitcoin suffer from, mainly in issues surrounding high computational inefficiency. While its efficient proof-of-stake system holds promise, security flaws raise concerns about patient data privacy. Using patient data as a mining incentive sparks ethical questions around consent and trust. However, the drop-dead difference in energy consumption between Ethereum and other blockchains, namely Bitcoin, addresses many environmental and data-colonialist social disparities linked to blockchain-based health records. Maintaining Ethereum's ethical and social edge over both other blockchain options and traditional, non-blockchain EHR systems is crucial to maintain confidence in developing a viable Ethereum-based EHR system, demanding innovative solutions for its integration in healthcare.

Index Terms—Ethereum, Blockchain, Proof of Work, Proof of Stake, Non-Maleficence, Consensus mechanisms, Blocks, Mining, Encryption.

I. INTRODUCTION — EXISTING LITERATURE AND ETHEREUM'S DIFFERENCES TO OTHER BLOCKCHAINS

As Ethereum continues to scale, its future ambitions branch away from an encapsulated financial asset to a generalized blockchain system that controls many non-financial aspects of future society. Examples of such include health records, real estate, and taxes, etc [1]. Blockchain and Ethereum can have big impacts in the future, as all big data-driven systems do. As Ethereum grows, an important question to consider is whether the values of Ethereum's creators cohere with those of their organization and of their clients [2]. It becomes increasingly important to bring about discussion on new ethical issues that are created from Ethereum's growth and increasing scale, including security risk related issues [3], environmental and energy consumption related issues [4], and human rights and data ethics — such as privacy, transparency and data ownership [5].

Existing research tends to look at how blockchain in general may impact medical and health records [5]. However, few connect these past examples with Ethereum. Ethereum has many major differences from other crypto systems, such as

proof of stake, which leads to much less power consumption and thus less damaging scaling of systems [6]. This leads to the possibility of many differences in the ethical and social issues that Ethereum may face when branching into medical records in the future.

This paper is an analysis of the said ethical issues that other blockchains may face when dealing with medical records — security, environment, human rights, and data ethics — pertain to Ethereum specifically. I aim to connect the existing literature and, using an analysis of the technical and related social and ethical differences between Ethereum and other blockchains regarding other domains, analyze them to paint a coherent picture of what ethical and societal effects Ethereum can have on the future of health records. I specifically aim to focus on a perspective of data colonialism: how blockchains exploit groups of people either to gather data or to grow and scale their systems.

II. EXISTING ETHICAL CONTEXT OF ELECTRONIC HEALTH RECORDS

A. Background — What an Electronic Health Record Is

The Office of the National Coordinator for Health Information Technology, a US governmental institution, offers a conclusive overview of what EHRs are:

An electronic health record (EHR) is a digital version of a patient's paper chart. EHRs are real-time, patient-centered records that make information available instantly and securely to authorized users. While an EHR does contain the medical and treatment histories of patients, an EHR system is built to go beyond standard clinical data collected in a provider's office and can be inclusive of a broader view of a patient's care. EHRs are a vital part of health IT and can contain a patient's medical history, diagnoses, medications, treatment plans, immunization dates, allergies, radiology images, and laboratory and test results, allow access to evidence-based tools that providers can use to make decisions about a patient's care, and automate and streamline provider workflow.

One of the key features of an EHR is that health information can be created and managed by authorized providers in a digital format capable of

being shared with other providers across more than one health care organization. EHRs are built to share information with other health care providers and organizations – such as laboratories, specialists, medical imaging facilities, pharmacies, emergency facilities, and school and workplace clinics – so they contain information from all clinicians involved in a patient’s care [7].

B. Benefits and Drawbacks of Current Systems

Existing electronic health record (EHR) systems are exceptionally fast and accurate at a low cost, which leads to less lost records, delays and healthcare fees, and an end result of improved patient compliance and quality assurance [8]. However, current EHR systems have long suffered with security vulnerabilities and breaches due to poor encryption, which leads to a failure in privacy and confidentiality, which are incredibly important in the domain of electronic health records and also legally required, dictated by HIPAA and HITECH legislation [9].

C. Privacy

Privacy is important morally and also as it upholds one of the well-defined four pillars of medical ethics which medical professionals abide by: non-maleficence, or doing no harm. Privacy is required to do no harm as it leads to effective communication between patient and medical caregiver and the patient feels less reluctant to give sensitive information that may save their life [10]. In essence, what is valued in a system of medical records are speed, accuracy and low cost, and especially a secure system that maintains patient confidentiality and privacy.



D. Caption: A display of the four pillars of medical ethics of which medical professionals abide by.

III. SECURITY AND DATA ETHICS

A systematic review of blockchain in personal health records observed that many studies that analyzed electronic health records (EHRs) and potential candidate blockchains to host them suggested Ethereum as the most optimal blockchain technology for EHRs [3]. According to the same systematic

review, however, many articles also proposed other more secure consensus mechanisms, and stated that most blockchain technologies, including ethereum, are not secure enough as of current consensus mechanisms because of methods such as inference/blockchain analysis [3]. This decreases the confidence that Ethereum is able to maintain patient privacy due to security breaches and means that in its current state, if Ethereum hosted medical records it would lead to privacy ethical complications. However, technologies can evolve, and the possibility of switching to a mechanism such as proof of authority — an option proposed by one of the reviewed studies — can be explored in the future [3].

A. Patient Data as a Mining Incentive

Yujin Han, et. al., also describes that utilizing Ethereum for medical records requires using patient data as an incentive for miners to mine blocks. This raises questions about informing patients about the use of their anonymized data and whether it should require patient consent [11], creating a data colonialism issue of how the blockchain could incorporate the data of the patients. Though the information would be encrypted and miners should be unable to access the patient data, it may prove to be anti-peace-of-mind for patients and could lead to their reluctance to receive medical aid [8].

IV. INEFFICIENCY, FAIRNESS AND TRUST

A. Inefficiency

One major difference between Ethereum and other blockchains is its proof of stake consensus mechanism, which makes it extremely energy-efficient compared to other blockchains. Ethereum did a merge in 2022 which switched from its defunct proof of work mechanism to this new system, which slashed energy requirements by 99.98% [6]. Yujin Han, et. al., suggests that one specific issue that blockchains must solve before becoming viable is blockchain inefficiency, and the subsequent high energy consumption required for consensus mechanisms [11]. However, the specific ethical issue that is high environmental impact from switching to a blockchain system for EHRs is minimal when we consider Ethereum’s low energy requirements due to proof of stake. In the context of data colonialism, Ethereum would take minimal advantage of marginalized communities in where the energy required to maintain the blockchain is gathered, versus other blockchains which utilize the significantly more energy-intensive proof of work mechanism.

B. Fairness

The described low-efficiency of other blockchains demand specialized hardware, potentially creating disparities favoring those with better resources. It could deepen existing inequalities in healthcare access, as those with better hardware and more energy resources gain more control within traditional proof of work blockchain-based healthcare systems [11]. This furthers the case for Ethereum as a host for EHR, because Ethereum’s lower energy requirements make it less prudent to own specialized hardware in order to operate any future Ethereum-based EHRs.

V. SPEED AND COST

A. Speed

Ethereum's 14 second block time is also extremely short compared to other blockchains. Bitcoin's, for example, is 10 minutes. Block time is a measure of how long it takes for the blockchain to create, verify and store a "block," or chunk of data in the blockchain. However, we must also consider block sizes. Ethereum's block sizes are in the ballpark of 100 kilobytes in size [12], while Bitcoin's is limited to 1 megabyte, or around 10 times the size of Ethereum's average block size [13].

Taking into account times per block and bytes per block, the Ethereum blockchain processes block data as such:

$$\frac{\left(\frac{100 \text{ kb}/ETH \text{ block}}{14 \text{ s/block}}\right)}{\left(\frac{1000 \text{ kb}/BTC \text{ block}}{(10 \text{ min/block})(3600 \text{ s/min})}\right)} = \frac{7.1429 \text{ kb/s on the ETH network}}{1.6667 \text{ kb/s on the BTC network}} = 4.2857$$

Thus the Ethereum network processes data 4.3 times faster than the Bitcoin network. Block speed is important when considering the amount of data a single medical patient generates into EHRs: 80 megabytes a year [14].

80,000 kb of data per year

$$\begin{array}{cc} \text{For ETH :} & \text{For BTC :} \\ \frac{80,000 \text{ kb}}{(7.1429 \text{ kb/s})(3600 \text{ s/h})} & \frac{80,000 \text{ kb}}{(1.6667 \text{ kb/s})(3600 \text{ s/h})} \end{array} \quad (1)$$

$$= 3.1111 \text{ hours} \quad = 13.3333 \text{ hours}$$

$$\Delta t = (13.3333 - 3.1111)(\text{hours per patient per year})$$

$$= 10.2222 \text{ hours per patient per year}$$

As pictured in the above derivation, EHRs stored on an Ethereum network would save 10.22 hours of processing time each year when compared to a network similar to Bitcoin. This is significant because medical conditions are dynamic and ever changing, and time saved processing the data means a higher chance the patient could be saved. For example, if a patient is being airlifted from one smaller hospital to another, medical records created at the smaller hospital must be processed in a timely manner to reach the larger hospital where the patient can receive more complex emergency treatment quickly. The increased speed of Ethereum and its subsequent chance to save more lives versus alternative blockchain EHR hosts such as Bitcoin touches upon the pillar of beneficence in the four medical ethics pillar: "doing good." This pillar encapsulates the idea that if something can be done to "do good," to save more lives, for example, then it should be adopted [15].

B. Cost

The cost to develop and operate electronic health record systems can vary greatly and specifically in the lens of data colonialism, be prohibitive against less wealthy regions. A blockchain-based EHR would require initial development costs and the costs to operate dApps*** and smart contracts. Estimates for developing a blockchain-based EHR system range from 400,000 to 1,500,000 [16], which may limit these systems to regions with good healthcare funding and already high technology. Though electricity costs vary, the average transaction on the Ethereum network with its proof-of-stake model ranges from 0.8 to 14.7 watt-hours [17], compared to Bitcoin's whopping 703,000+ wh per transaction [18].

Ethereum would be much less data colonialist versus proof-of-work based blockchains when hosting EHR not only due to the decreased environmental impact but the decreased cost for the medical professionals to implement the system, increasing accessibility to more regions. Additionally, the lower power cost of Ethereum versus other blockchains makes it more affordable to the end users, which are the medical patients. Specifically in countries without standardized, free, government-subsidized healthcare, such as in the United States, it could lead to decreased medical insurance costs and uninsured medical bills. This could be realized in a cause and effect relationship: a higher likelihood for a patient to receive treatment, and thus fulfilling the second pillar of Medical Ethics [15].

VI. CONCLUSION

Ethereum holds potential for transforming healthcare systems with its efficient proof-of-stake mechanism and touted advantages, though security vulnerabilities raise ethical dilemmas, potentially compromising patient data privacy. Utilizing patient data as an incentive for mining raises questions about patient consent and may impact the trust between those seeking medical help and medical providers. However, Ethereum's energy efficiency is exemplary versus other blockchains, addressing concerns about environmental impacts and social disparities that studies relate to blockchain-based EHRs in general. Ethereum is also significantly faster and cost-effective at validating blocks, thus having the chance of saving more lives due to speed and a lessened cost burden. Condensed, the effects of Ethereum's better efficiency, open source workings, and high speed and low cost advance data decolonization despite its shortcomings. Balancing Ethereum's benefits with these ethical and social issues and solving such issues through innovation (i.e. a further improved consensus mechanism) is critical for ensuring its integration into healthcare aligns with ethical values and what is important in an electronic medical record.

ACKNOWLEDGEMENT

This paper contains content generated by ChatGPT which has been revised and fact-checked.

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